In [ ]:

```
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#Registration No: 240905630
#Section: PG
#Roll No: 34
```

In [2]:

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
```

In [7]:

```
df = pd.read_csv('sentiment.csv')
df
```

Out[7]:

	ID	Text	Sentiment	Timestamp	User	Platform	Hashtags
0	0	Enjoying a beautiful day at the park! ...	Positive	15-01-2023 12:30	User123	Twitter	#Nature #Park
1	1	Traffic was terrible this morning. ...	Negative	15-01-2023 08:45	CommuterX	Twitter	#Traffic #Morning
2	2	Just finished an amazing workout! 💪 ...	Positive	15-01-2023 15:45	FitnessFan	Instagram	#Fitness #Workout
3	3	Excited about the upcoming weekend getaway! ...	Positive	15-01-2023 18:20	AdventureX	Facebook	#Travel #Adventure
4	4	Trying out a new recipe for dinner tonight. ...	Neutral	15-01-2023 19:55	ChefCook	Instagram	#Cooking #Food
...	...	...	...	...	...	...	...
727	728	Collaborating on a science project that receiv...	Happy	18-08-2017 18:20	ScienceProjectSuccessHighSchool	Facebook	#ScienceFairWinner #HighSchoolScience
728	729	Attending a surprise birthday party organized ...	Happy	22-06-2018 14:15	BirthdayPartyJoyHighSchool	Instagram	#SurpriseCelebration #HighSchoolFriendship
729	730	Successfully fundraising for a school charity ...	Happy	05-04-2019 17:30	CharityFundraisingTriumphHighSchool	Twitter	#CommunityGiving #HighSchoolPhilanthropy
730	731	Participating in a multicultural festival, cel...	Happy	29-02-2020 20:45	MulticulturalFestivalJoyHighSchool	Facebook	#CulturalCelebration #HighSchoolUnity
731	732	Organizing a virtual talent show during challe...	Happy	15-11-2020 15:15	VirtualTalentShowSuccessHighSchool	Instagram	#VirtualEntertainment #HighSchoolPositivity

732 rows × 14 columns

In [73]:

```
#1. In the sentiment feature and text, trim white spaces and convert to lowercase
df['Sentiment'] = df['Sentiment'].str.strip().str.lower()
df['Text'] = df['Text'].str.strip().str.lower()
df
```

Out[73]:

	ID	Text	Sentiment	Timestamp	User	Platform	Hashtags	I
0	0	enjoying a beautiful day at the park!	positive	15-01-2023 12:30	User123	Twitter	#Nature #Park	
1	1	traffic was terrible this morning.	negative	15-01-2023 08:45	CommuterX	Twitter	#Traffic #Morning	
2	2	just finished an amazing workout! 💪	positive	15-01-2023 15:45	FitnessFan	Instagram	#Fitness #Workout	
3	3	excited about the upcoming weekend getaway!	positive	15-01-2023 18:20	AdventureX	Facebook	#Travel #Adventure	
4	4	trying out a new recipe for dinner tonight.	neutral	15-01-2023 19:55	ChefCook	Instagram	#Cooking #Food	
...	...	...	...	...	...	...	...	
727	728	collaborating on a science project that receiv...	happy	18-08-2017 18:20	ScienceProjectSuccessHighSchool	Facebook	#ScienceFairWinner #HighSchoolScience	
728	729	attending a surprise birthday party organized ...	happy	22-06-2018 14:15	BirthdayPartyJoyHighSchool	Instagram	#SurpriseCelebration #HighSchoolFriendship	
729	730	successfully fundraising for a school charity ...	happy	05-04-2019 17:30	CharityFundraisingTriumphHighSchool	Twitter	#CommunityGiving #HighSchoolPhilanthropy	
730	731	participating in a multicultural festival, cel...	happy	29-02-2020 20:45	MulticulturalFestivalJoyHighSchool	Facebook	#CulturalCelebration #HighSchoolUnity	
731	732	organizing a virtual talent show during challe...	happy	15-11-2020 15:15	VirtualTalentShowSuccessHighSchool	Instagram	#VirtualEntertainment #HighSchoolPositivity	

732 rows × 16 columns

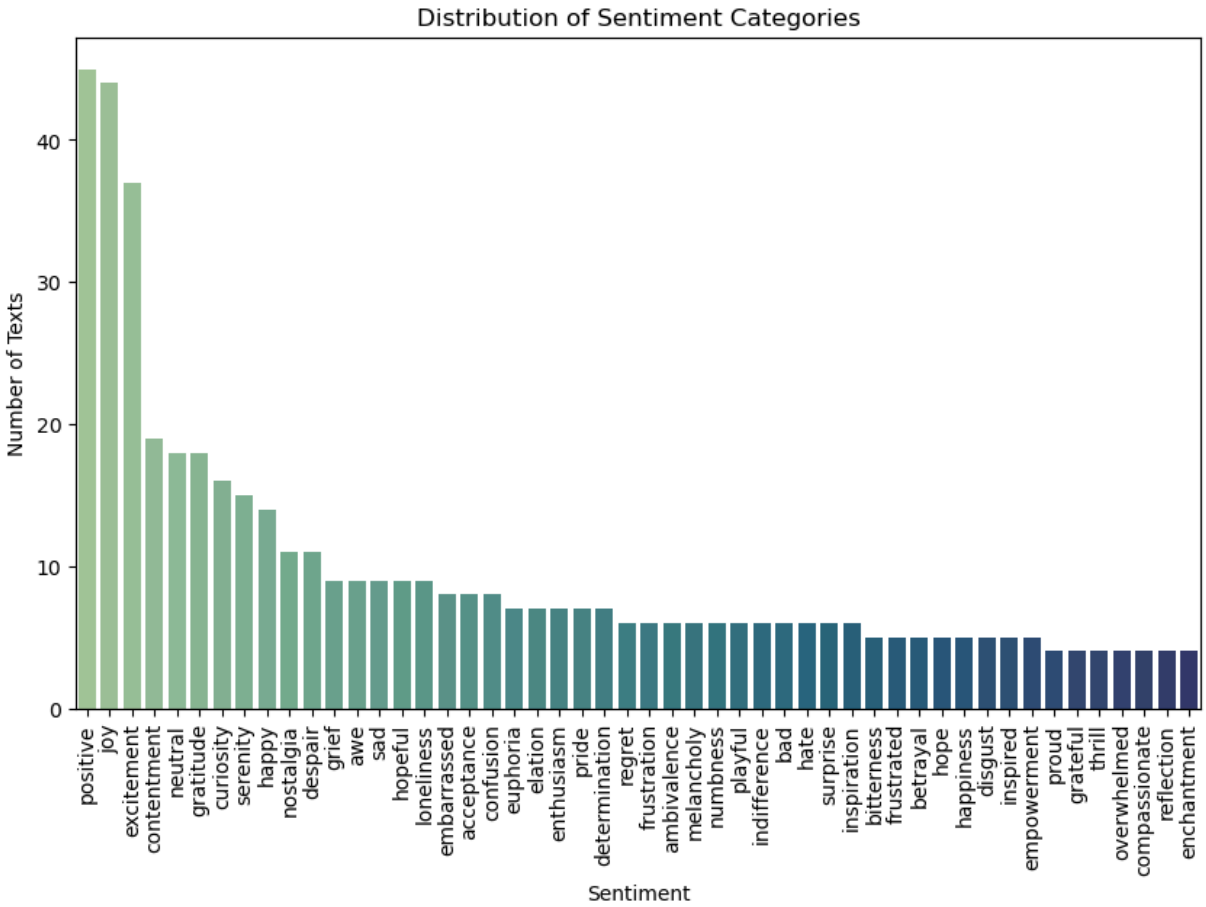
In [20]:

```python
#2. What are the sentiment categories? Tabulate and visualize the number of texts for each
sentiment_counts = df['Sentiment'].value_counts()

print("\nSentiment Categories and Counts:")
print(sentiment_counts.head(50))

plt.figure(figsize=(10, 6))
sns.barplot(data=df, x=sentiment_counts.head(50).index, y=sentiment_counts.head(50).values
plt.title('Distribution of Sentiment Categories')
plt.xlabel('Sentiment')
plt.ylabel('Number of Texts')
plt.xticks(rotation=90)
plt.show()
```

```
Sentiment Categories and Counts:
positive         45
joy              44
excitement       37
contentment      19
neutral          18
gratitude        18
curiosity        16
serenity         15
happy            14
nostalgia        11
despair          11
grief             9
awe               9
sad               9
hopeful           9
loneliness        9
embarrassed       8
acceptance        8
confusion         8
euphoria          7
elation           7
enthusiasm        7
pride             7
determination     7
regret            6
frustration       6
ambivalence       6
melancholy        6
numbness          6
playful           6
indifference      6
bad               6
hate              6
surprise          6
inspiration       6
bitterness        5
frustrated        5
betrayal          5
hope              5
happiness         5
disgust           5
inspired          5
empowerment       5
proud             4
grateful          4
thrill            4
overwhelmed       4
compassionate     4
reflection        4
enchantment       4
Name: Sentiment, dtype: int64
```

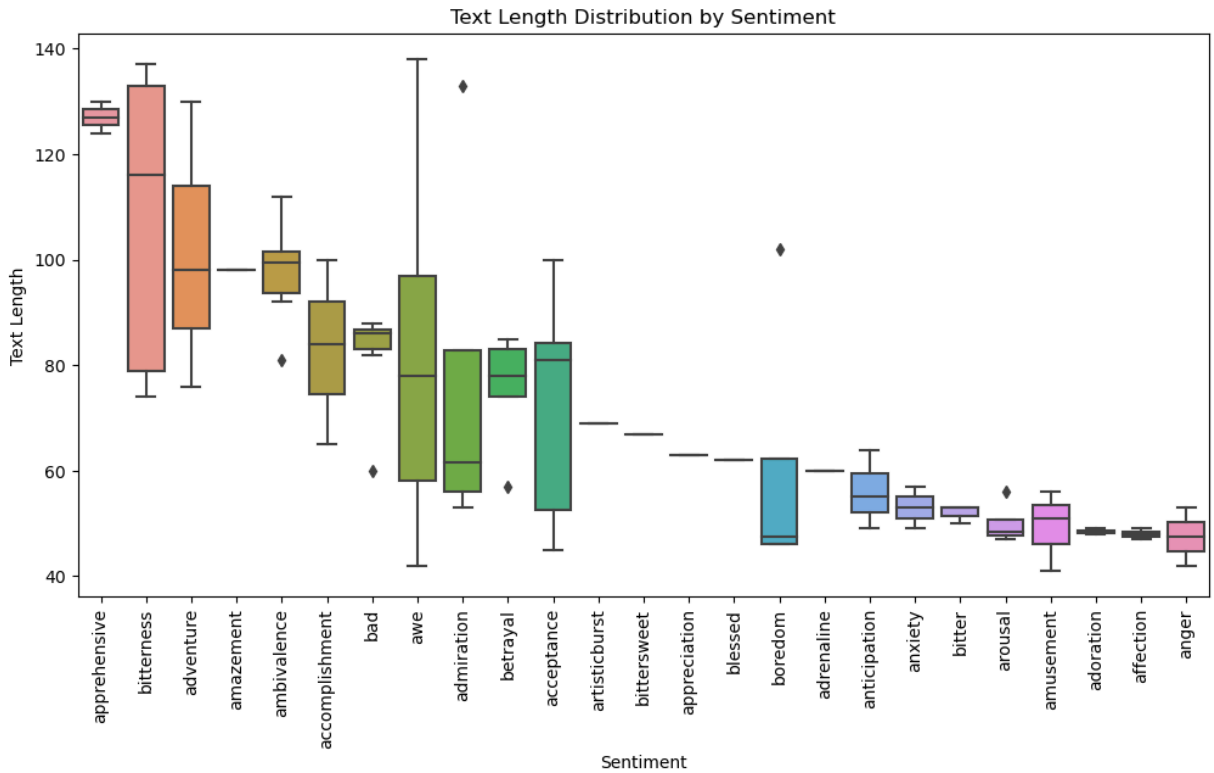

Distribution of Sentiment Categories
Number of Texts
0
10
20
30
40
positive
joy
excitement
contentment
neutral
gratitude
curiosity
serenity
happy
nostalgia
despair
grief
awe
sad
hopeful
loneliness
embarrassed
acceptance
confusion
euphoria
elation
enthusiasm
pride
determination
regret
frustration
ambivalence
melancholy
numbness
playful
indifference
bad
hate
surprise
inspiration
bitterness
frustrated
betrayal
hope
happiness
disgust
inspired
empowerment
proud
grateful
thrill
overwhelmed
compassionate
reflection
enchantment
Sentiment

In [74]:

```
#3. Tabulate and Visualize the text length for each sentiment categor
df['Text_Length'] = df['Text'].apply(len)
sentiment_length = df.groupby('Sentiment')['Text_Length'].mean().head(25).sort_values(asce
print("\nAverage Text Length by Sentiment (Top 25):")
print(sentiment_length)

plt.figure(figsize=(12, 6))
sns.boxplot(data=df, x='Sentiment', y='Text_Length', order=sentiment_length.index)
plt.title('Text Length Distribution by Sentiment')
plt.xlabel('Sentiment')
plt.ylabel('Text Length')
plt.xticks(rotation=90)
plt.show()
```

```
Average Text Length by Sentiment (Top 25):
Sentiment
apprehensive      127.000000
bitterness        107.800000
adventure         101.333333
amazement          98.000000
ambivalence        97.666667
accomplishment     83.000000
bad                81.500000
awe                80.555556
admiration         77.250000
betrayal           75.400000
acceptance         72.500000
artisticburst      69.000000
bittersweet        67.000000
appreciation       63.000000
blessed            62.000000
boredom            60.750000
adrenaline         60.000000
anticipation       56.000000
anxiety            53.000000
bitter             52.000000
arousal            50.000000
amusement          49.333333
adoration          48.500000
affection          48.000000
anger              47.500000
Name: Text_Length, dtype: float64
```



In [61]:

```python
#4. Find and display the text(s) with the maximum number of likes
max_likes = df['Likes'].max()
top_liked_texts = df[df['Likes'] == max_likes][['Text', 'Sentiment']]
print(f"\nText(s) with Maximum Likes ({max_likes}):")
top_liked_texts
```

```
Text(s) with Maximum Likes (80):
```

Out[61]:

	Text	Sentiment
335	thrilled to witness the grandeur of a cultural...	thrill
345	motivated to achieve fitness goals after an in...	motivation
355	anticipation for an upcoming adventure in an e...	anticipation
368	elation over discovering a rare book in a quai...	elation
382	a sense of wonder at the vastness of the cosmo...	wonder
402	awe-inspired by the vastness of the cosmos on ...	wonder
432	heartache deepens, a solitary journey through ...	despair
470	dancing on sunshine, each step a celebration o...	joy
481	surrounded by the colors of joy, a canvas pain...	joy
510	at the front row of adele's concert, each note...	emotion
520	at a justin bieber concert, the infectious bea...	enthusiasm
530	captivated by the spellbinding plot twists, th...	excitement
540	celebrating a historic victory in the world cu...	joy
550	after a series of defeats, the soccer team fac...	disappointment
560	in the serene beauty of a sunset, nature unfol...	tranquility
570	underneath the city lights, the dancer express...	mesmerizing

In [60]:

```python
#5. Find and display the text(s) with the maximum reposts. what is its sentiment?
max_reposts = df['Retweets'].max()
top_reposted_texts = df[df['Retweets'] == max_reposts][['Text', 'Sentiment']]
print(f"\nText(s) with Maximum Reposts ({max_reposts} reposts):")
print(top_reposted_texts)
```

```
Text(s) with Maximum Reposts (40 reposts):
                                                Text       Sentiment
335  thrilled to witness the grandeur of a cultural...          thrill
345  motivated to achieve fitness goals after an in...      motivation
355  anticipation for an upcoming adventure in an e...    anticipation
368  elation over discovering a rare book in a quai...         elation
382  a sense of wonder at the vastness of the cosmo...          wonder
402  awe-inspired by the vastness of the cosmos on ...          wonder
432  heartache deepens, a solitary journey through ...         despair
470  dancing on sunshine, each step a celebration o...             joy
481  surrounded by the colors of joy, a canvas pain...             joy
510  at the front row of adele's concert, each note...         emotion
520  at a justin bieber concert, the infectious bea...      enthusiasm
530  captivated by the spellbinding plot twists, th...      excitement
540  celebrating a historic victory in the world cu...             joy
550  after a series of defeats, the soccer team fac...  disappointment
560  in the serene beauty of a sunset, nature unfol...     tranquility
570  underneath the city lights, the dancer express...     mesmerizing
```

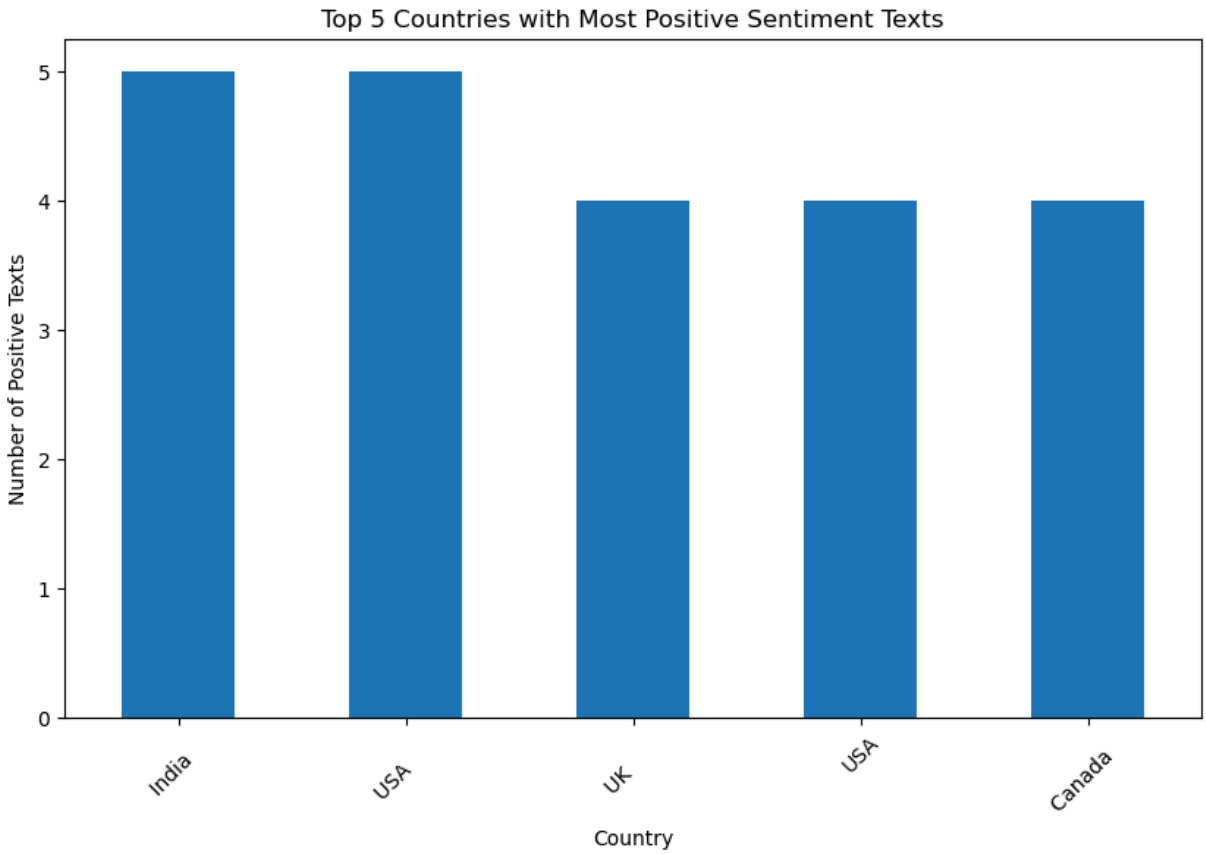
In [63]:

```python
#6. Tabulate and Visualize the top 5 countries that have posted the most texts with a posi

positive_countries = df[df['Sentiment'] == 'positive'].groupby('Country').size()
top_positive_countries = positive_countries.sort_values(ascending=False).head(5)
print("\nTop 5 Countries with Most Positive Sentiment Texts:")
print(top_positive_countries)

plt.figure(figsize=(10, 6))
top_positive_countries.plot(kind='bar')
plt.title('Top 5 Countries with Most Positive Sentiment Texts')
plt.xlabel('Country')
plt.ylabel('Number of Positive Texts')
plt.xticks(rotation=45)
plt.show()
```

```
Top 5 Countries with Most Positive Sentiment Texts:
Country
 India        5
 USA          5
 UK           4
USA           4
 Canada       4
dtype: int64
```



In [66]:

```python
#7. Find and display all the text(s) with the word ‘Travel’ in their hashtag
travel_texts = df[df['Hashtags'].str.contains('travel', case=False, na=False)][['Text']]

print("\nTexts with 'Travel' in Hashtags:")
print(travel_texts)
```

```
Texts with 'Travel' in Hashtags:
                                                Text
3          excited about the upcoming weekend getaway!
72   excitement building up for the upcoming vacation!
84   enjoying every moment of this trip—pure enjoym...
145  arousal of excitement before a much-awaited trip.
596  spontaneously booked a weekend getaway. advent...
610  embarked on a road trip to revisit cherished p...
622  embarked on a solo travel adventure, discoveri...
```

In [71]:

```python
#8. Find and display the most popular (most number of likes) posts (top 5) from the year 2
negative_2023 = df[(df['Year'] == 2023) & (df['Sentiment'] == 'negative')]
top_negative_2023 = negative_2023.nlargest(5, 'Likes')[['Text', 'Likes', 'User', 'Timestam

print("\nTop 5 Most Liked Negative Sentiment Posts from 2023:")
print(top_negative_2023)
```

```
Top 5 Most Liked Negative Sentiment Posts from 2021:
                                                Text  Likes            User  \
8   political discussions heating up on the timeline.     60   DebateTalk
11             feeling a bit under the weather today.     15   WellnessCheck
19                   winter blues got me feeling low.     15   WinterBlues
1                  traffic was terrible this morning.     10   CommuterX

           Timestamp
8   17-01-2023 08:00
11  18-01-2023 10:30
19  20-01-2023 15:15
1   15-01-2023 08:45
```

In [75]:

```python
#9. Find and display a list of Users and their posts whose texts are labeled ‘Happiness’ s
happiness_posts = df[df['Sentiment'] == 'happiness'][['User', 'Text', 'Likes', 'Timestamp'

print("\nUsers and Posts with 'Happiness' Sentiment:")
print(happiness_posts)
```

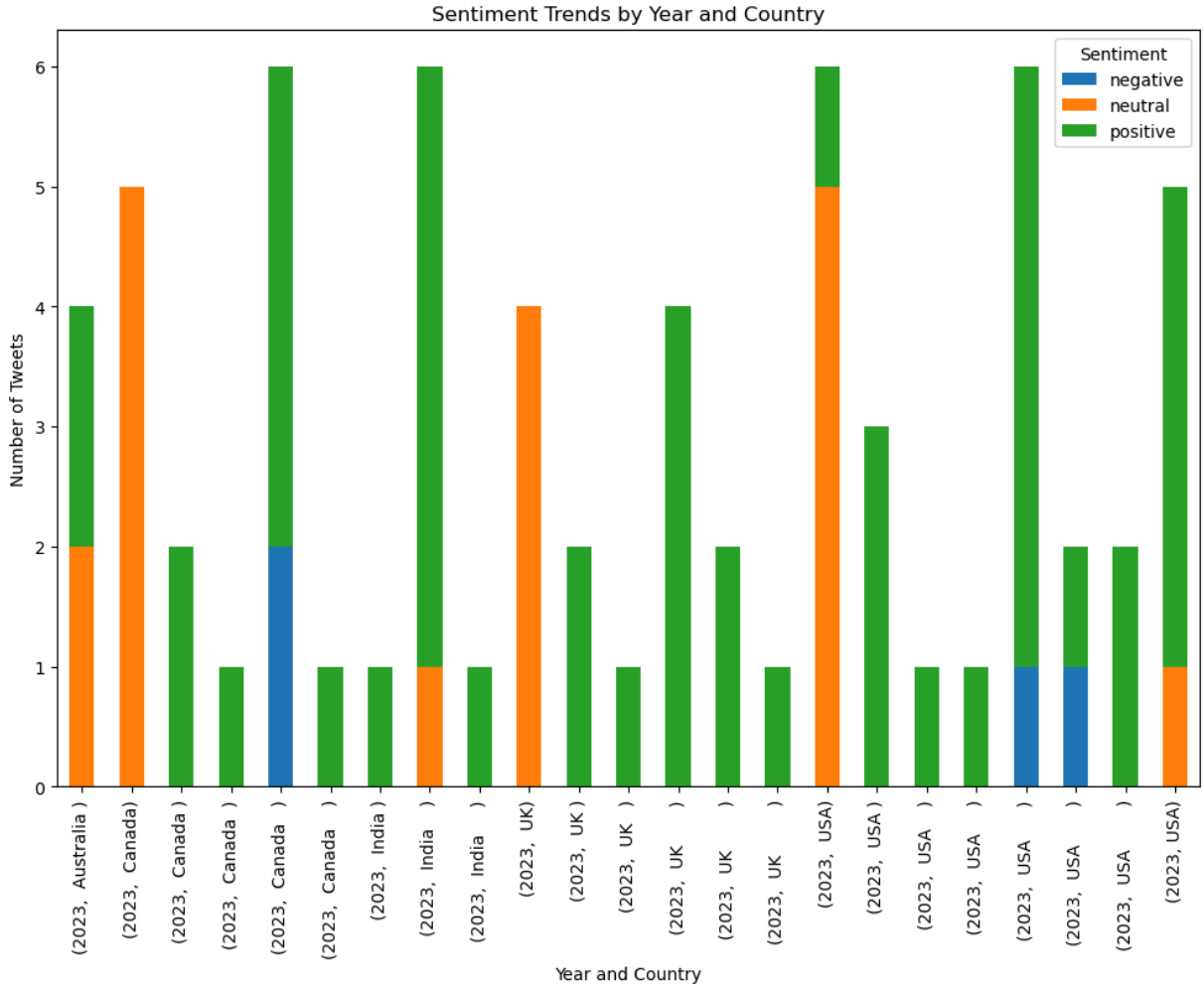
```
Users and Posts with 'Happiness' Sentiment:
                     User                                              Text  \
56       ProudFriend       pure happiness: celebrating a loved one's achi...
80       NewParentJoy      overflowing happiness: welcoming a new family ...
367   SunnyDayEnthusiast   happiness blooms like flowers in a garden on a...
471       EchoedLaughter   laughter echoes in the air, a chorus of happin...
488     LaughterSymphony   a symphony of laughter, each note a key to unl...

     Likes         Timestamp
56      60  13-02-2023 10:00
80      60  22-02-2023 10:00
367     30  10-02-2020 11:15
471     70  12-04-2018 18:30
488     60  02-08-2021 19:00
```

In [80]:

```
#10. Tabulate and visualize the years and countrywide tweets of the sentiments. Pick tweet
selected_sentiments = ['positive', 'negative', 'neutral']
filtered_df = df[df['Sentiment'].isin(selected_sentiments)]
sentiment_by_year_country = filtered_df.groupby(['Year', 'Country', 'Sentiment']).size().u
print(sentiment_by_year_country)
sentiment_by_year_country.plot(kind='bar', stacked=True, figsize=(12, 8))
plt.title('Sentiment Trends by Year and Country')
plt.xlabel('Year and Country')
plt.ylabel('Number of Tweets')
plt.xticks(rotation=90)
plt.show()
```

```
Sentiment          negative  neutral  positive
Year Country
2023  Australia         0.0      2.0       2.0
      Canada            0.0      5.0       0.0
      Canada            0.0      0.0       2.0
      Canada            0.0      0.0       1.0
      Canada            2.0      0.0       4.0
      Canada            0.0      0.0       1.0
      India             0.0      0.0       1.0
      India             0.0      1.0       5.0
      India             0.0      0.0       1.0
      UK                0.0      4.0       0.0
      UK                0.0      0.0       2.0
      UK                0.0      0.0       1.0
      UK                0.0      0.0       4.0
      UK                0.0      0.0       2.0
      UK                0.0      0.0       1.0
      USA               0.0      5.0       1.0
      USA               0.0      0.0       3.0
      USA               0.0      0.0       1.0
      USA               0.0      0.0       1.0
      USA               1.0      0.0       5.0
      USA               1.0      0.0       1.0
      USA               0.0      0.0       2.0
     USA                0.0      1.0       4.0
```


Sentiment Trends by Year and Country
Sentiment
negative
neutral
positive
Number of Tweets
0
1
2
3
4
5
6
(2023, Australia)
(2023, Canada)
(2023, Canada)
(2023, Canada)
(2023, Canada)
(2023, Canada)
(2023, India)
(2023, India)
(2023, India)
(2023, UK)
(2023, UK)
(2023, UK)
(2023, UK)
(2023, UK)
(2023, UK)
(2023, USA)
(2023, USA)
(2023, USA)
(2023, USA)
(2023, USA)
(2023, USA)
(2023, USA)
(2023, USA)
Year and Country